# HOURHOUSE

*Adult ESL · Newark, NJ*

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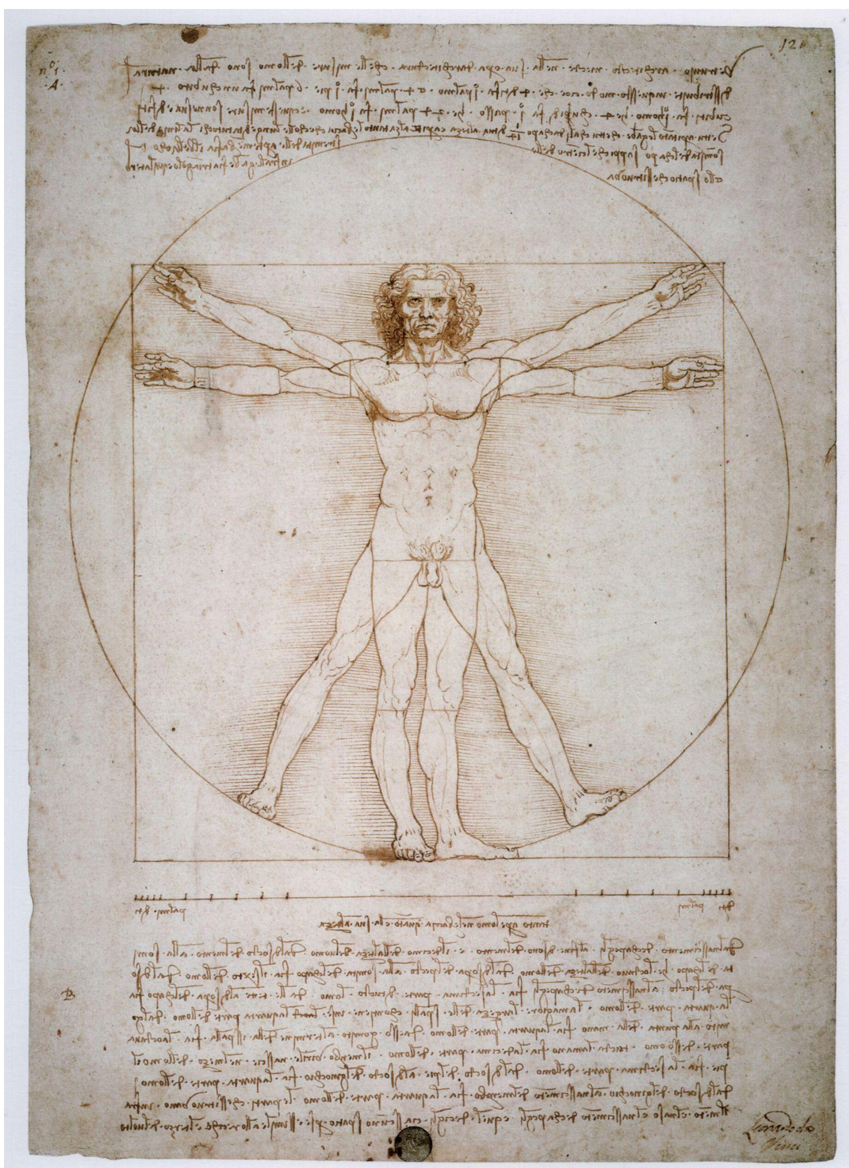
**ACADEMIC ENGLISH · CAMBRIDGE B2 / C1**

# The Vitruvian Approximation

*From the Rhind Papyrus to Lean 4*

*A four-session workbook*

**1 hour · Monday–Thursday · 4 hours total**



# About this workbook

| | |
|---|---|
| **Source Text** | The Vitruvian Approximation: Rational Circle-to-Square Transitions in the dm³ Framework — Pablo Nogueira Grossi (2026). An accessible rewrite for B2–C1 students appears in Session 2. |
| **Video** | The Beautiful Story Behind Da Vinci's Vitruvian Man (16 min 29 sec) — [youtube.com/watch?v=MHd8C8X0BlM](https://youtube.com/watch?v=MHd8C8X0BlM) |
| **Level** | Upper-Intermediate to Advanced (B2–C1) |
| **Format** | Four 60-minute sessions, one per day, Monday–Thursday |
| **Skills covered** | Listening · Reading · Speaking · Writing · Vocabulary · Grammar · Critical Thinking |
| **Topic** | Mathematics · History of Science · Renaissance Art · Computer-Assisted Proof · Academic English |
| **Course objective** | By the end of the four sessions, students will be able to discuss the history of squaring the circle, read and interpret an accessible academic article written by their teacher, follow a short Lean 4 proof of the Rhind result, and produce a written and spoken response in academic English. |
| **Materials** | This workbook (printed, one per student)<br>Whiteboard / markers · pens · highlighters<br>QR code (page 3) for the video<br>Optional: laptop or projector to display Lean 4 code in Session 3 |

**Scan to watch the video**

*The Beautiful Story Behind Da Vinci's Vitruvian Man — 16 min 29 sec*

[youtube.com/watch?v=MHd8C8X0BlM](https://youtube.com/watch?v=MHd8C8X0BlM)

# Course timing at a glance

| Session | Day | Focus | Time |
|---|---|---|---|
| 1 | Monday | From Vitruvius to Newark — Video, Vocabulary, Skills focus | 60 min |
| 2 | Tuesday | Reading the Paper, Honestly — Accessible text + comprehension | 60 min |
| 3 | Wednesday | Squaring the Circle in Lean 4 — Listening, Code reading, Grammar | 60 min |
| 4 | Thursday | Your Voice, Your Approximation — Speaking, Writing, Application | 60 min |
| **TOTAL** | | | **240 min** |

*Each session contains four to six exercises mapped to the CAELA stages: Warm-Up, Presentation, Guided Practice, Communicative Practice, Evaluation and Application.*

SESSION 1 · Monday · 60 minutes

# From Vitruvius to Newark

**Today you will:**

- watch a short documentary about Leonardo's Vitruvian Man
- discuss what circles, squares and proportion mean to you
- learn 12 academic words from the paper you will read tomorrow
- practise sentence-linking — a key Cambridge B2/C1 reading skill

## 1 Warm-up *(10 min)*

**In pairs,** discuss the following questions. Be ready to share one answer with the class.

1. Look at a circle and a square. What does each shape make you think of? Choose three words for each.
2. Have you ever heard of the number π (pi)? What do you know about it? Where do we use it in everyday life?
3. The word approximation means a value that is close to but not exactly correct. Give two examples of approximations from everyday life — for example, in cooking, in time, or in money.
4. Today's reading paper opens with this dedication: "Neither of them read from a recipe book. Both of them ate the cake. This is what education is, and what it is not." What do you think this means?

## 2 Listening preview — Video viewing *(12 min)*

**Before you watch,** scan the QR code on page 3 or follow the link. While you watch the first 8 minutes, take notes to answer these questions:

1. Who was Marcus Vitruvius Pollio?
2. In what century did he live?
3. What did Vitruvius believe about the proportions of the human body?
4. Why did Renaissance artists like Leonardo become interested in his ideas?
5. What does the word "proportion" mean? Try to explain it in your own words.

**After the video:** compare your answers with a partner. Be ready to add one detail you remember that your partner did not write down.

## 3 Vocabulary focus — Part A *(8 min)*

**Complete the words below by adding the missing vowels** *(a, e, i, o, u)*. All twelve words appear in tomorrow's reading.

1. `___ pp r ___ x ___ m ___ t ___ ___ n` (n.) — a value close to, but not exactly equal to, the correct one
2. `r ___ t ___ ___ n ___ l` (adj.) — able to be written as a fraction p / q with whole numbers
3. `tr ___ nsc ___ nd ___ nt ___ l` (adj.) — describes a number that is not the solution of any algebraic equation
4. `c ___ nv ___ rg ___ nt` (n.) — a fraction in a sequence that gets closer and closer to a target value
5. `pr ___ p ___ rt ___ ___ n` (n.) — a balanced relationship between parts
6. `h ___ rm ___ ny` (n.) — a pleasing, balanced arrangement

**7.** ___ p ___ r ___ t ___ r (n.) — a mathematical rule that transforms an input into an output

**8.** c ___ nj ___ ct ___ r ___ (n.) — an idea believed to be true but not yet proven

**9.** fr ___ m ___ w ___ rk (n.) — a structured system of rules used to analyse a problem

**10.** f ___ rm ___ l ___ z ___ (v.) — to write something precisely using strict logic or code

**11.** h ___ dg ___ ng (n.) — careful or cautious language used when you are not 100% sure (e.g., "it may be that…")

**12.** scr ___ b ___ (n.) — a person in ancient times whose job was to write

## 4 Vocabulary focus — Part B *(10 min)*

**Now use the complete words from Part A** to fill the gaps in the following sentences. Use each word once.

**1.** The fraction 22/7 is a well-known ____________________ for the value of π.

**2.** Mathematicians have proven that π is ____________________ , which means no polynomial equation with whole-number coefficients can produce it.

**3.** A _______________ number can always be written as p divided by q, where p and q are integers.

**4.** The continued-fraction expansion of π produces ____________________ s like 333/106 and 355/113 that get closer and closer to the true value.

**5.** Vitruvius believed that _______________ in architecture should reflect the _______________ of the human body.

**6.** A ____________________ is an idea that seems very likely to be true but has not yet been fully proven.

**7.** The G-cycle is an _______________ chain that applies a series of transformations to circle data.

**8.** The team used Lean 4 to _______________ the theorem and make it verifiable by computer.

**9.** The dm³ _______________ provides a structured set of rules for modelling contact dynamics.

**10.** In academic writing we use _______________ — phrases like "it appears that" — to avoid sounding too certain.

**11.** The Egyptian _______________ Ahmes wrote down the rule about 3,650 years ago.

## 5 Skills focus — sentence linking *(12 min)*

**In Cambridge Reading Part 6/7 tasks** you must show that one sentence connects to the previous one. Read sentences 1–8 and match them to options a–h. Then *underline* the words in a–h that link to the previous sentence.

**1.** Vitruvius believed that the human body was a model of cosmic proportion.

**2.** The Rhind Papyrus is one of the oldest mathematical documents we know.

**3.** In the 19th century, Lindemann proved that π is transcendental.

**4.** Leonardo drew the Vitruvian Man around 1490.

**5.** Lean 4 is a programming language designed for writing mathematical proofs.

**6.** Pablo's paper makes a careful, honest claim about what its framework can and cannot do.

**7.** The compression ratio 8/9 is admissible inside the dm³ basin of attraction.

**8.** Pi cannot be written as an exact fraction, no matter how many digits we use.

**Options:**

**a.** This means that no straightedge-and-compass construction can ever produce a length equal to √π.

**b.** Yet, useful approximations such as 22/7 and 355/113 are good enough for almost every practical purpose.

**c.** For this reason, his ideas were re-read enthusiastically by Renaissance painters and architects.

**d.** It was made on papyrus around 1650 BCE by an Egyptian scribe called Ahmes.

**e.** The drawing is now kept in Venice and remains one of the most famous images in the history of science.

**f.** This means that other compression ratios near 8/9 are also acceptable, but the framework alone does not select 8/9 uniquely.

**g.** Such proofs can then be checked automatically by a computer.

**h.** In other words, it does not pretend to do more than it actually does.

## 6 Wrap-up — exit ticket *(8 min)*

**Write 3–4 sentences** answering this prompt. Hand in to the teacher before you leave.

*Prompt:* Which of the twelve vocabulary words from today is the most useful for you, and why? Use the word in a sentence about your own work, study or life.

SESSION 2 · Tuesday · 60 minutes

# Reading the Paper, Honestly

**Today you will:**

- read an accessible version of your teacher's research paper
- fill 10 gaps using the words you learned yesterday
- answer comprehension questions in academic English
- discuss the idea of “honest incompleteness” as a class

## 1 Warm-up — pair recall *(5 min)*

**Without looking at your notes,** tell your partner three things you remember about the video and four words from yesterday's vocabulary list. Then check page 3 — were you correct?

## 2 Pre-reading prediction *(5 min)*

**Before you read,** look at the title and the section headings on the next page and predict the answers to these questions:

1. How old is the oldest written approximation of π that we know?
2. Does the paper claim to solve the problem of squaring the circle?
3. What does the symbol “8/9” mean in this paper?
4. In one sentence, what do you think “honest incompleteness” means?

## 3 Reading — fill the gaps *(25 min)*

**Read the article carefully.** Then fill the ten numbered gaps with the words from the box. Each word is used once. There are two extra words that you do not need.

**Word box (12 words, 10 are used):**

- admits · approximations · basin · conjectures · conditions · convergents · framework · leads · partly · rational · transcendental · two

## The Vitruvian Approximation

*Adapted from the original paper by Pablo Nogueira Grossi (2026) — accessible version for B2–C1 students.*

*Original DOI: 10.5281/zenodo.19984522*

**0 Dedication.** To David and Giulia, my great gurus. Neither of them read from a recipe book. Both of them ate the cake. This is what education is, and what it is not.

**1 Two faces of an old problem.** The problem of “squaring the circle” — drawing a square with the same area as a given circle — has 1 ____________ different faces. The *classical face* asks for an *exact* construction with only a compass and a straightedge. In 1882, Lindemann proved that π is *transcendental*, which means that π is not the solution of any algebraic equation. As a result, no exact construction is possible. The *practical face* is different. It asks for a 2 ____________ approximation that is good enough for a particular purpose. People have been finding such approximations for thousands of years.

**2 The oldest answer we know.** The earliest written record is from the Rhind Mathematical Papyrus, written in Egypt around 1650 BCE. The *scribe* Ahmes wrote a rule that, in modern words, says: “A circle of diameter 9 has roughly the same area as a square of side 8." This 3 ____________ to π ≈ 256/81 ≈ 3.1605, which is accurate to about 0.6% of the true value of π.

**3 What this paper does — and what it does not do.** This paper takes a modern mathematical 4 ____________ called Generative Contact Mechanics (GCM) and asks: "Does our framework rediscover the ancient Egyptian rule on its own?" The answer is honest: 5 ____________. We can write the operator chain so that one application of it produces the Rhind result by direct calculation. But the ratio 8/9 is built into the operator from the start; the framework does not generate the ratio out of nothing.

**4 The G-cycle in plain words.** Imagine a circle described by two pieces of rational data: its diameter and its area. The G-cycle is a sequence of four steps:

1. **Compression (C).** The diameter is shrunk by a factor of 8/9.
2. **Kinking (K).** The new diameter is treated as the side of a square — four right angles break the circular symmetry.
3. **Folding (F).** The side length is squared to produce an area.
4. **Unfolding (U).** The square area is released as the final output.

The result is 6 ____________ : start with a circle of diameter d and you finish with a square of area $(8d/9)^2$. For d = 9 you get 64. This is the Rhind rule.

**5 Stability — what dm³ does, and what it doesn't.** The dm³ system has three numerical 7 ____________ : an exponential stability rate, a Gronwall radius, and an embodiment threshold. Together they describe when a small disturbance returns to equilibrium and when it doesn't. In the rational squaring setting, these conditions are happy with the 8/9 compression ratio: the system stays inside its 8 ____________ of attraction. But many other ratios near 8/9 would also be acceptable. So dm³ *confirms* the Rhind rule, but does not *derive* it. This is an honest, open problem.

**6 A selection functional, and a confessed conjecture.** The paper proposes a simple measurement called $\mathscr{C}$_rat. It multiplies the approximation error |p/q − π| by the complexity of the fraction (log p + log q). Good fractions are accurate *and* simple. The *Main Conjecture* states that the minimisers of $\mathscr{C}$_rat are precisely the continued-fraction 9 ____________ of π — fractions like 22/7, 333/106 and 355/113. The author 10 ____________ openly says that he cannot fully prove this yet, and so the conjecture is marked `sorry` in the Lean 4 file. This is not a failure: it is part of a tradition the paper calls "Honest Incompleteness".

**7 Why Lean4?** Lean 4 is a programming language for writing mathematical proofs that a computer can check. The paper includes a small Lean file in which the Rhind theorem is fully proved by a computer (we will read it together in Session 3). Conjecture 5.4 is included too, but it is left open with the keyword `sorry`, so that any reader can see exactly what has been proven and what has not.

**8 Honest Incompleteness.** Many academic papers use very confident language. This paper does the opposite. It uses *hedging* — phrases like "it appears that" and "the answer is: partly" — to make clear what is solid, what is suggestive, and what remains open. The aim is to show students of mathematics, and students of English, that intellectual honesty is itself a form of clarity.

**9 Conclusion.** Squaring the circle is impossible by the rules of compass and straightedge. But asking for a *good rational approximation* is a different and answerable question. Ancient Egyptians answered it 3,650 years ago; Archimedes refined it; Renaissance artists drew it; Lean 4 can now check it. The aim of this paper — and of these four lessons — is to show that learning is also an approximation: we get closer to truth one careful step at a time.

## 4 Reading comprehension — Part 5 style *(15 min)*

**Choose the best answer** (A, B, C or D) for each question. Be ready to defend your answer with evidence from the text.

**1.** Why is exact squaring of the circle impossible with compass and straightedge?
- **A.** Because nobody has tried hard enough.
- **B.** Because π is transcendental and $\sqrt{\pi}$ is not constructible.
- **C.** Because Egyptians did not have compasses.
- **D.** Because Lean 4 has not yet proved it.

**2.** What does the author mean when he says the answer to his main question is “partly”?
- **A.** The framework recovers the Rhind rule as a calculation but does not derive 8/9 from scratch.
- **B.** The framework is wrong about half of the time.
- **C.** Only some readers will understand it.
- **D.** The paper is unfinished.

**3.** What does “honest incompleteness” refer to in this paper?
- **A.** A weakness in the author's mathematics.
- **B.** A practice of clearly marking which results are proved and which are conjectured.
- **C.** Forgetting to finish the references.
- **D.** A new theorem about prime numbers.

**4.** Why does the author keep the conjecture marked `sorry` inside the Lean 4 file?
- **A.** To apologise to the reader.
- **B.** Because Lean 4 cannot read English.
- **C.** So that any reader can see exactly what is proved and what is not.
- **D.** Because the conjecture has been disproved.

**5.** Which best summarises the author's view of education?
- **A.** Following recipes carefully.
- **B.** Memorising as many fractions as possible.
- **C.** Eating the cake — getting closer to the truth one careful step at a time.
- **D.** Avoiding mistakes at all costs.

## 5 Discussion — “Honest Incompleteness” *(10 min)*

**In groups of three,** discuss for 7 minutes, then share with the class.

**1.** Why might it be useful — for a scientist, a teacher or a student — to clearly say “I don't know yet”?

**2.** Have you ever pretended to know something you did not? What happened? What would have happened if you had said “sorry, I don't know”?

**3.** How is “honest incompleteness” different from being lazy or careless? Try to give an example.

SESSION 3 · Wednesday · 60 minutes

# Squaring the Circle in Lean 4

**Today you will:**

- practise Cambridge-style listening with three short audios
- read a real Lean 4 proof, line by line
- complete the missing words inside the proof
- practise the passive voice and academic hedging

## 1 Warm-up — Lean as a language *(5 min)*

**In pairs,** discuss:

1. If a programming language can "speak" mathematics, in what way is it like a human language? In what way is it different?
2. Why might a mathematician want a computer to check her proof?

## 2 Listening — Cambridge Part 1 style *(20 min)*

**You will hear three short audios.** Your teacher will read each one aloud (or play a recording). Listen and complete the notes. You will hear each audio twice.

### Audio 1 — Voicemail from a colleague at a maths conference *(four gaps)*

**FROM:** Anna · **TO:** Pablo

**Message:**

Anna says she is unlikely to make today's session because she has been delayed in 1 ____________________.

She would like to discuss the 8/9 compression with Pablo, especially the question of whether 2 ____________________ ratios near 8/9 are also basin-admissible.

She suggests they meet at 3 ____________________ tomorrow morning, and asks Pablo to bring the printed copy of the 4 ____________________ paper.

### Audio 2 — Short interview with the author of the paper *(four gaps)*

**Notes from the interview:**

- The paper does not claim to solve the classical squaring problem; it works in the 5 ____________________ approximation regime.
- The author wrote the paper while teaching adult ESL students in 6 ____________________ , New Jersey.
- Conjecture 5.4 is left open and marked 7 ____________________ in the Lean file.
- The author hopes the paper will help English students see that 8 ____________________ honesty is a form of clarity, not a weakness.

### Audio 3 — Two students after the lesson *(four gaps)*

**Marisol:** “So at the start I was worried because the title says 9 ____________________ but my partner said ‘we are going to read it like a story, not solve it'.”

**Andrés:** “Right. And the part I liked best was the line: 10 ____________________ of them ate the cake. I never thought a maths paper could start with a line like that.”

**Marisol:** “What surprised me most was that the author 11 ____________________ openly that one part is not yet proven. I thought professors only wrote about what they had finished.”

**Andrés:** “And that the language used by Lean is the same kind of 12 ____________________ that we are learning — names, definitions, careful steps.”

## 3 Reading Lean 4 — fill the gaps *(20 min)*

**Below is the Lean 4 file from the appendix of the paper, with seven gaps.** Use the words in the box to complete the proof. Three words are extra.

**Word box:**

- diameter · ring · theorem · structure · sorry · compress · fold · import · trivial · open

```
-- Lean 4 — Squaring the Circle (rational instance)
-- File: AXLE/Vitruvian.lean

___________   Mathlib.Data.Rat.Basic
import Mathlib.Tactic

/-- A rational circle has a diameter and an area, both rational. -/
___________   RationalCircle where
      ___________   : ℚ
      area     : ℚ

def C___________    (c : RationalCircle) : RationalCircle :=
      { diameter := c.diameter * 8 / 9, area := c.area }

def K_kink (c : RationalCircle) : ℚ := c.diameter
def F___________    (side : ℚ) : ℚ := side ^ 2
def U_unfold (a : ℚ) : ℚ := a

def G_rat (c : RationalCircle) : ℚ :=
    U_unfold (F_fold (K_kink (C_compress c)))

/-- Theorem 3.4: For diameter 9, the G-cycle returns 64. -/
___________    rhind_recovery :
      G_rat { diameter := 9, area := 64 } = 64 := by
    simp [G_rat, C_compress, K_kink, F_fold, U_unfold]
    ring

/-- Conjecture 5.4 — left open in the tradition of honest incompleteness. -/
theorem G_rat_selects_CF_convergents : True := by
    ___________
```

**Answer questions about the code:**

1. Which line of the code corresponds to step C — Compression — of the G-cycle? Line ____
2. Which line corresponds to step F — Folding? Line ____
3. Which lines together form the full G-cycle G_rat? Lines ____ to ____
4. In line 15–18, which line is the actual proof step that finishes the calculation? Line ____
5. Why is line 21 left as it is, instead of being a finished proof? Use one full sentence.

## 4 Grammar focus — passive voice + academic hedging *(15 min)*

**A. Rewrite each sentence in the passive voice.** Keep the meaning the same.

1. Ahmes wrote the rule on the Rhind Papyrus around 1650 BCE.
2. Lindemann proved the transcendence of π in 1882.
3. Researchers formalised the theorem using Lean 4.
4. Da Vinci drew the Vitruvian Man around 1490.
5. The author has not yet proved Conjecture 5.4.

**B. Rewrite each sentence using academic** *hedging*. Choose from: *it appears that, the evidence suggests that, it may be that, in the rational regime, this paper does not claim to.*

1. The framework solves the squaring of the circle.
2. 8/9 is the only correct compression ratio.
3. The conjecture is true.
4. The paper proves more than it actually proves.

SESSION 4 · Thursday · 60 minutes

# Your Voice, Your Approximation

**Today you will:**

- present a topic for one minute, then answer a partner's question
- plan and write a 100-word academic paragraph
- reflect on what you have learned across all four sessions
- receive your portfolio mark and feedback

## 1 Warm-up — recall in pairs *(5 min)*

**Without your workbook,** each partner names:

- one person from the video (Session 1)
- one section from the paper (Session 2)
- one Lean keyword from the code (Session 3)
- one example of academic hedging (any session)

## 2 Speaking — Cambridge Part 2 style *(20 min)*

**Work in pairs.** Each student picks one card. You have **1 minute** to prepare and **1 minute** to speak. Then your partner asks one follow-up question. Then swap roles.

**CARD 1 Discovery vs. Recovery**

The paper says the G-cycle "packages the known ratio in operator language; it does not generate 8/9 from a blank circle." Talk for one minute about the difference between discovering something new and recovering something already known. Use one example from your own learning.

**CARD 2 Honest Incompleteness**

The author marks one of his theorems as "sorry" — meaning he believes it is true but cannot yet prove it. Talk for one minute about why intellectual honesty matters in academic work. Should students do the same in essays and exams?

**CARD 3 Ancient Wisdom, Modern Math**

The Rhind Papyrus is nearly 3,700 years old, yet a paper published in 2026 in Newark still refers to it. Talk for one minute about a piece of knowledge from your own culture or country that is very old but still useful today.

**CARD 4 Newark to the World**

The paper was written by your teacher, in Newark, NJ, and published as an international preprint. Talk for one minute about what it means to you that serious academic work can come from your own community. Has knowing the author changed how you read the paper?

**CARD 5 Computers and Truth**

Lean 4 lets a computer check a mathematical proof. Talk for one minute about whether a computer can really “verify truth”, or whether truth always needs a human reader. Use one example from outside mathematics.

**CARD 6 Approximation as a Way of Living**

Pi cannot be written exactly. We use approximations because they are good enough. Talk for one minute about an area of life — relationships, work, language — where a “good enough” answer is more useful than a perfect one that does not exist.

## 3 Writing — academic paragraph *(25 min)*

**Plan (5 min):** Choose ONE prompt. Write a small plan: thesis sentence, two supporting points with evidence, concluding sentence. Then write a 100-word paragraph (15 min) and check it for hedging and passive voice (5 min).

**PROMPT A Summarise**

In one paragraph (≈100 words), summarise The Vitruvian Approximation in your own words. Use at least six vocabulary words from Session 1 and at least one example of academic hedging.

**PROMPT B Personal Response**

Pick the idea from these four sessions — from the video, the paper, the Lean code, or the discussions — that most surprised, moved or changed you. In ≈100 words, explain what it was and why it matters to you. Use at least one passive sentence and one hedge.

**PROMPT C Honest Incompleteness in Your Field**

Where in your own work, study or daily life would “honest incompleteness” — clearly stating what you know and what you don't — make things better? In ≈100 words, give one concrete example. Use at least three vocabulary words from this workbook.

**Your paragraph:**

## 4 Application + Closing reflection *(10 min)*

**Pair share, then class share.** Answer in 3–5 sentences each.

**1.** Think of something important you learned by doing rather than by following instructions. What was it? How did it feel to figure it out yourself?

**2.** π cannot be written exactly. Where else in life is a “good enough” answer more useful than a perfect one that does not exist?

**3.** These four sessions connect ancient Egypt, Renaissance Italy and modern Newark. Where do you come from, and what knowledge did you bring with you into this room?

*Closing line — read aloud together: “Every approximation gets us closer to the truth. So does every lesson.”*

# TEACHER MATERIALS

*Answer keys, listening transcripts, and lesson timing notes for the four-session course.*

## Course Timing Summary

| Session | Day | Focus | Time |
|---|---|---|---|
| 1 | Monday | From Vitruvius to Newark — Video, Vocabulary, Skills focus | 60 min |
| 2 | Tuesday | Reading the Paper, Honestly — Accessible text + comprehension | 60 min |
| 3 | Wednesday | Squaring the Circle in Lean 4 — Listening, Code reading, Grammar | 60 min |
| 4 | Thursday | Your Voice, Your Approximation — Speaking, Writing, Application | 60 min |
| **TOTAL** | | | **240 min** |

## Session 1 — Answer Key

### Exercise 3 — Vocabulary Part A (vowels filled in)

- approximation
- rational
- transcendental
- convergent
- proportion
- harmony
- operator
- conjecture
- framework
- formalize
- hedging
- scribe

### Exercise 4 — Vocabulary Part B

1. approximation
2. transcendental
3. rational
4. convergent
5. proportion / harmony
6. conjecture
7. operator
8. formalize
9. framework

10. hedging
11. scribe

## Exercise 5 — Sentence linking

1 → c · 2 → d · 3 → a · 4 → e · 5 → g · 6 → h · 7 → f · 8 → b

*Linking words to underline:*

- c. “For this reason” refers back to Vitruvius's belief about cosmic proportion.
- d. “It was made” refers back to the Rhind Papyrus.
- a. “This means that” refers to the transcendence of π.
- e. “The drawing” refers to Leonardo's Vitruvian Man.
- g. “Such proofs” refers to mathematical proofs in Lean 4.
- h. “In other words” rephrases the careful, honest claim.
- f. “This means that other compression ratios” refers to the 8/9 admissibility.
- b. “Yet, useful approximations” contrasts with the impossibility of exact fraction.

## Session 2 — Answer Key

### Exercise 3 — Reading: gap fills

1 = two · 2 = rational · 3 = leads · 4 = framework · 5 = partly · 6 = approximations · 7 = conditions · 8 = basin · 9 = convergents · 10 = admits

*Extra words (not used): conjectures, transcendental.*

### Exercise 4 — Comprehension (Cambridge Part 5 style)

1. B · 2. A · 3. B · 4. C · 5. C

### Exercise 5 — Discussion (sample teacher answers)

- Q1 — Saying “I don't know yet” protects credibility, invites collaboration, and prevents misinformation. In Lean 4, the keyword sorry serves the same role — it marks an honest gap that other mathematicians can finish.
- Q2 — Personal answers vary; encourage students to use academic hedging when generalising from their experience.
- Q3 — Honest incompleteness is active: it requires the speaker to identify exactly what is missing. Laziness or carelessness is silent. Example: a Lean file with sorry is more useful than a file that pretends to be complete.

## Session 3 — Listening Transcripts + Answer Key

### Audio 1 — Voicemail from a colleague (read by teacher)

*“Hi Pablo, it's Anna. There's been a bit of a setback on the journey — I've been delayed in Newark airport because of weather, and I don't think I'll make today's session. Could we meet at Bob's Café tomorrow morning at 10:30 instead? I really want to talk about the 8/9 compression and whether other compression ratios near 8/9 are also basin-admissible. Could you bring your printed copy of the Vitruvian paper? Cheers.”*

**Answers:** 1 = Newark airport · 2 = other / different · 3 = Bob's Café · 4 = Vitruvian

### Audio 2 — Short interview with the author

*Interviewer: “Does your paper claim to solve the squaring of the circle?”Author: “No. It doesn't claim to solve the classical problem. It works entirely in the rational approximation regime. I wrote the paper in Newark, New Jersey, while teaching adult ESL students. Conjecture 5.4 is left open and clearly marked sorry in the Lean file. I hope readers, especially English students, will see that intellectual honesty is a form of clarity, not a weakness.”*

**Answers:** 5 = rational · 6 = Newark · 7 = sorry · 8 = intellectual

### Audio 3 — Two students after the lesson

*Marisol: “So at the start I was worried because the title says squaring the circle but my partner said ‘we are going to read it like a story, not solve it.'”Andrés: “Right. And the part I liked best was the line: Both of them ate the cake. I never thought a maths paper could start with a line like that.”Marisol: “What surprised me most was that the author admits openly that one part is not yet proven. I thought professors only wrote about what they had finished.”Andrés: “And that the language used by Lean is the same kind of English that we are learning — names, definitions, careful steps.”*

**Answers:** 9 = squaring the circle · 10 = Both · 11 = admits · 12 = English

### Exercise 3 — Lean 4 gaps

Gaps in order: import · structure · diameter · compress · fold · theorem · sorry

**Code questions:**

- Q1 — Compression: line 7–8 (def C_compress …)
- Q2 — Folding: line 10 (def F_fold …)
- Q3 — Full G-cycle: lines 12–13 (def G_rat …)
- Q4 — Actual finishing step: line 18 (ring)
- Q5 — Line 21 is left as “sorry” because the conjecture has not yet been proven; the author marks the gap honestly so any reader can see it.

### Exercise 4A — Passive voice (model answers)

1. The rule was written on the Rhind Papyrus by Ahmes around 1650 BCE.
2. The transcendence of π was proved by Lindemann in 1882.
3. The theorem was formalised using Lean 4.
4. The Vitruvian Man was drawn by Da Vinci around 1490.
5. Conjecture 5.4 has not yet been proved by the author.

### Exercise 4B — Hedging (model answers)

1. This paper does not claim to solve the squaring of the circle.
2. It may be that 8/9 is not the only acceptable compression ratio.
3. The evidence suggests that the conjecture is true.
4. It appears that the paper proves less than its title suggests.

## Session 4 — Marking Guide

### Exercise 2 — Speaking (Cambridge B2/C1 rubric)

- Grammatical Range and Accuracy (5) — variety of structures, including passive voice and hedging.
- Lexical Resource (5) — at least three workbook vocabulary items used naturally.
- Discourse Management (5) — clear structure: thesis, support, example, conclusion.
- Pronunciation (5) — academic stress on key words: ap-PROX-i-MA-tion, tran-SCEN-den-tal, con-VER-gent.
- Interactive Communication (5) — answers the partner's question fully and asks a relevant question back.
- Total: ____ / 25

### Exercise 3 — Writing (Cambridge B2/C1 rubric)

- Content (5) — addresses prompt fully, ≈100 words.
- Communicative Achievement (5) — appropriate academic register, evidence of hedging.
- Organisation (5) — thesis sentence, support, conclusion.
- Language (5) — accurate use of passive voice and at least 3 workbook vocabulary items.
- Total: ____ / 20

## Lesson Notes — for the coordinator

- This four-session unit has been designed to satisfy CAELA's stage-sequence model (Warm-Up → Presentation → Guided Practice → Communicative Practice → Evaluation → Application) by mapping the seven CAELA stages across the four 60-minute days rather than within a single block.
- For multilevel classes, use mixed-level pairs in Sessions 1, 2 and 4 (fluency and content) and same-level pairs in Session 3 (accuracy with passive voice and Lean syntax).
- If a class is closer to B2 than C1, simplify the Lean reading by reading lines aloud as a class and underlining only the keywords (import, structure, def, theorem, sorry).
- If a class is closer to C1 than B2, extend Session 4 with a short oral defence: each student delivers their paragraph in 90 seconds with two follow-up questions from the class.
- For faculty observation: this workbook can also serve as a model lesson plan during teacher training. The CAELA evaluation rubric maps directly onto Sessions 2 (Presentation), 3 (Guided/Communicative Practice), and 4 (Application).
- The QR code on page 3 always links to the same YouTube video. Print one copy per student; one master copy with answers stays with the teacher.

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*Workbook designed using the CAELA Framework for Effective Lesson Planning for Adult English Language Learners.*